Language over demo breadcrumbs

Frozen LeWorldModel on Push-T

Alexander Bashirov Didar Seyidov Faith Ogunwoye

Skoltech

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The problem

LeWorldModel encodes a Push-T frame as 192 numbers and predicts the next latent from an action. The public checkpoint stays frozen. It was not trained with language.
The planner reaches a goal image about 25 demonstration steps ahead: 45/50 on the 4090, 47/50 on the 5090 setup check.
It does not reach a long live goal (0/20 at budget 50), and it does not spin the T in place. A rendered in-place goal scores 0.00 at budgets 50 and 100.
The course task is to let a sentence name a goal this planner can follow.

What we tried first

Angle is already in the latent. A ridge readout has held-out circular error 12.08 degrees on 180° frames and 14.58 degrees near angle 0. The encoder is not finetuned.
A constant latent step per command does not fit. Residual ratio 0.991.
A planning cost written on the angle probe reaches +90° on 0.10 of 20 starts and 180° on 0.00. Three rendered stages at +30°, +60°, and +90° pass the in-place gate on 1 of 20.
The miss is the planner under this horizon, not a missing text encoder.

The idea

The planner is reliable for a short hop. Expert demonstrations already contain those hops.

Build a graph on the train split: one node every 5 demonstration steps. Forward edges follow a demo. Jump edges link different demos that pass near the same state.

From the live state, take the cheapest route and hand the planner the frame about 25 steps along it. Execute, read the graph again, and repeat. Words choose the goal region. The graph chooses the frames.

Demo breadcrumbs

Task	n	Flat	Crumbs
Dataset goal, 50 steps ahead	50	24	34
Dataset goal, 100 steps ahead	50	10	29
Live random goal	20	0	3
Seven word templates	64	18	34

Budget 300. Flat is the same planner aimed at the final frame.

The dry graph has a route for 50/50 offset-50 tasks and for 13/20 live resets. On the live set, median block error falls from 162 px to 58 px.

Words as the goal

Seven templates: on the target, turn 90 degrees either way, push left, right, up, or down. Qwen2.5-1.5B-Instruct, local and greedy, maps a sentence to one command. The test set is 120 sentences, written before the parser was scored.

	Count	Rate
Parser correct	101/120	0.842
Keyword baseline	31/120	0.258
Gold command executed	57/111	0.514
Parsed command executed	51/111	0.459

Nine pushes are excluded because the gold destination leaves the arena. Parsing costs 6 successes. Execution is the larger gap.

Turns are not in-place

On the 20 turn templates, crumbs reach 8/20 and flat reaches 4/20. The success rule allows 40 px of drift.
The stricter gate, drift under 15 px, is 0/20 for crumbs and 1/20 for flat. The block does turn, and it slides while it turns.
The demonstrations do contain in-place turns: 8,040 of 427,396 nodes, in 1,691 of 16,817 episodes. Having the frame in the dataset is not the same as the short-horizon planner following it.

A head instead of the graph

A small network predicts the next latent from the start latent, the current latent, and a frozen sentence embedding. At test time there is no graph and no parser.
Validation error is 0.67 times the error of copying the current latent.
On the same 111 sentences the head reaches 18/111. With the sentence replaced by another row's text, it reaches 10/111. The word effect is on "put it on the target" (11/30 against 2/30), not on turns or pushes.
Parser plus graph remains at 51/111.

What the defense should take

Short demonstration goals are followable. Long goals and in-place spins are not, with this horizon and budget.

Routes through other demonstrations raise success from 24/50 to 34/50 at offset 50, and from 10/50 to 29/50 at offset 100.

A sentence can name the route. The 1.5B parser is right on 101/120 lines, and the parsed run loses 6 successes against the gold command.

Predicting the latent from the sentence, without retrieving a demonstration frame, is the weaker interface.